QMOC-SVM on Raspberry Pi 5 / OP-TEE — Self-Contained Report

This summarizes the work in **pi5-optee** against Cultice et al., *Anomaly Detection for Real-World Cyber-Physical Security using Quantum Hybrid Support Vector Machines* (arXiv:2409.04935). Primary sources: `docs/hai-qhsvm/hai-qhsvm-pi5-tee-qmoc-svm.md`, the HAI tables / resource docs, and `docs/experimental-data/*.csv`.

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1. Methodology

1.1 Problem and pipeline

The task is **anomaly detection** on the **HAI 20.07** ICS dataset (normal vs attack), using a **hybrid quantum–classical one-class SVM**:

- **Classical preprocessing** (host), following the paper's §IV-A: 60-sample moving average; categorical → frequency encoding; z-score; decision-tree Gini feature ranking; selected features mapped to rotation angles.
- **Quantum fidelity kernel** (evaluated inside an **OP-TEE Trusted Application** on the Pi 5 as a **classical statevector simulation**—no QPU, no GPU): dense-angle encoding (2 features per qubit: RY/RZ); nearest-neighbour CNOT ring + further rotations; Hilbert–Schmidt fidelity $K = |\langle\phi(x)|\phi(y)\rangle|^2$; post-process $k\langle = e\langle$ (self-similarity → e).
- **Classical One-Class SVM** (host, scikit-learn `OneClassSVM`, precomputed kernel) on those kernel values.

1.2 Shared evaluation draw

All models below are scored on the **same 600-row draw** (seed **2409**): **400 normal + 200 attack**. That is a **2:1** mix (as in the paper's Fig. 5 style), **not** the natural ~**3.94%** attack rate of the full HAI test files. Positive class = attack.

2. What QMOC-SVM changes (variations)

QOC-SVM here means a paper-faithful (or near-faithful) fidelity one-class run. **QMOC-SVM** is the robustness-tuned variant used when the paper-style kernel collapses.

Setting	QOC-SVM (paper-like)	QMOC-SVM
TA command	TA_QAI_CMD_QOC_BLOCK	TA_QAI_CMD_QMOC_BLOCK
Entangling-layer repetitions	3	1
Angle scale	full [–π, π]	data-driven bandwidth (×0.18 @ 8q; ×0.12 @ 16q)
Normal landmarks	200 (8q) / 16 (16q cut)	1,000 (8q) / 32 (16q cut)
nu	swept; labels often unchanged	chosen on 400 held-out normals (~5% FAR target)

Bandwidth selection (no test labels): pick the scale whose off-diagonal median fidelity lies in [0.1, 0.5] and is closest to **0.30**.

- 8q: **0.18** → median fidelity **0.280** (48 landmarks); **0.273** on full 1000×1000 Gram
- 16q: **0.12** → median **0.413** (16 landmarks); **0.399** on 32×32 Gram

nu selection: closest calibration false-alarm rate to **5%**.

- 8q: **nu=0.01** → FAR **6.5%**, **55** support vectors
- 16q: no nu near 5% (FAR ~**63%** for all); report **nu=0.01**, **18** SVs

2.1 Variants actually measured

Variant	Width	Notes
TinyML	16 features	Supervised logistic in TA (sees attack labels)
QMOC-SVM 8q	8q / 16 feat	Main one-class result
QOC-SVM 8q	8q / 16 feat	Paper norm [–π, π], 3 reps, 200 refs

Variant	Width	Notes
QOC-SVM 8q ×0.1	8q	Sensitivity check (not paper norm)
Classical RBF OC-SVM	16 feat	Host baseline, same columns before angle map
QMOC-SVM 16q	16q / 32 feat	Same circuit family; landmarks cut for runtime
QOC-SVM 16q	16q / 32 feat	Paper-like collapse check

3. Experimental setup (Raspberry Pi 5 + OP-TEE)

Item	Value
Board	Raspberry Pi 5 Model B Rev 1.1
CPU	4× Cortex-A76 @ 2.4 GHz
Normal-World RAM	~2,021,376 kB
Kernel	Linux 6.18.46-v8-16k+
Trust environment	Real OP-TEE TA (CFG_TA_FLOAT_SUPPORT=y)
TA UUID	24173bc2-5143-4d91-92c5-ffae31dc618d
Timing	TEE_GetSystemTime inside the TA
Quantum / GPU	none — statevector on one secure Cortex-A76 core
Host SVM	scikit-learn on Normal World

Memory footprint (illustrative): 8q signed `.ta` ~**85 KiB**, ~**137 KiB** reserved per call; 16q enlarges `.bss` to ~**1.62 MB** for a 65,536-amplitude statevector. Gram / calibration / test matrices stay on the **host** (QMOC 8q scoring held ~**8.1 MB** of matrices; post-fit only **55×64 B = 3,520 B** of SVs).

Dataset: HAI 20.07 from `icsdataset/hai`; after MA60 ≈ **550,682** train / **444,482** test rows; attack rate ≈ **3.94%**.

4. Main results (shared 600-row draw)

4.1 Classification

Method	Acc	Prec	Rec	F1
TinyML (supervised, TA)	84.0%	0.736	0.810	0.771
QMOC-SVM 8q	66.2%	0.495	0.710	0.583
QOC-SVM 8q ×0.1	49.3%	0.384	0.865	0.532
Classical RBF OC-SVM	47.2%	0.373	0.860	0.520
QMOC-SVM 16q	38.5%	0.346	0.950	0.507
QOC-SVM 8q, paper norm	24.0%	0.265	0.720	0.387
QOC-SVM 16q, paper norm	23.2%	0.258	0.695	0.376
Paper Table I quantum (published)	87%	0.88	0.87	0.86
Paper Table I classical (published)	73%	0.81	0.73	0.65

QMOC-SVM 8q confusion: TN **255**, FP **145**, FN **58**, TP **142**. **Paper-like QOC-SVM 8q:** TN **0**, FP **400** — every normal flagged as attack (fidelity collapse: distinct rows nearly orthogonal at full [–π, π] with 3 reps).

Takeaways on this draw

- QMOC-SVM 8q is the **strongest one-class TA result** (24% → **66.2%** Acc; F1 0.387 → **0.583** vs collapsed QOC).
- It beats classical RBF OC-SVM here (**47.2%** / F1 **0.520**).
- It remains behind supervised TinyML (**84%** / F1 **0.771**).
- It **does not reproduce** Table I's **87% / F1 0.86**.

4.2 TEE timing (illustrative)

Kernel work	Pairs	TEE time	ms/pair
QMOC 8q Gram+cal+test	2,000,000	169.4 s	0.085
QOC 8q Gram+test	160,000	39.5 s	0.247
QMOC 16q Gram+cal+test	33,024	1,689 s	51.1

One QMOC 8q test row vs 1,000 landmarks ≈ **84.7 ms** TA time; one QMOC 16q row vs 32 landmarks ≈ **1.64 s**. Full-file paper-scale kernels are estimated as multi-year work on one secure core (documented resource analysis).

5. How this differs from arXiv:2409.04935

Paper (Cultice, Onim, Giani, Thapliyal): Qiskit **noiseless simulation** of a fidelity one-class SVM (**QOC-SVM**): MA60 + z-score + Gini selection; dense-angle map; feature map **repeated 3×**; angles in **[–π, π]**; k■ = e■; classical OC-SVM. Headline: **87% Acc / F1 0.86** at 16 features / 8 qubits (~14 pp above their classical SVM at 73%).

Dimension	Paper	This work
Execution	IBM Qiskit sim	OP-TEE TA statevector on Pi 5
Trust / deployment	None	Kernel in Secure World; SVM on host
Primary reported recipe	3 reps, [–π, π]	QMOC: 1 rep + bandwidth + more landmarks + cal-nu
Literal paper recipe on Pi	—	Collapses to ~24% Acc (TN=0)
Best one-class result here	—	QMOC 8q 66.2% / F1 0.583
Table I reproduced?	Claims 87% / 0.86	No
Extra baselines	Classical SVM in paper tables	Same-draw TinyML, RBF OC-SVM, 8q/16q QOC & QMOC
Resource measurement	Conceptual NISQ cost	Measured ms/pair, TA memory, full-file projections
Unpublished in paper (called out here)	Train/test widths, nu, exact gate listing beyond prose	Explicit subsets, hyperparams, and TEE numbers

Interpretation in the repo: at the paper's stated full-angle, 3-rep schedule, typical distinct HAI rows become nearly orthogonal, so OC-SVM rejects almost everything. QMOC restores a **graded** kernel and lifts one-class performance substantially, but still does not validate Table I under the published normalization/circuit description as implemented in the TA. Remaining gap vs 87% is attributed in part to unpublished sample sizes / nu / possible milder feature-map details—not solely "quantum vs classical."

6. Bottom line

QMOC-SVM is a **TEE-deployable, hyperparameter-stabilized** fidelity one-class SVM in the Cultice et al. circuit family: same Hilbert–Schmidt / e■ pipeline, but **one repetition**, **training-pair bandwidth**, **more landmarks**, and **calibration-chosen nu**. On Pi 5 OP-TEE it clearly outperforms a naive paper-faithful QOC-SVM (**66% vs 24%**), edges classical RBF on this split, stays behind supervised TinyML (**84%**), and **does not confirm** the paper's **87% / F1 0.86** HAI claim under the published angle and repetition settings as run here.

Key write-ups: `docs/hai-qhsvm/hai-qhsvm-pi5-tee-qmoc-svm.md`, `docs/hai-qhsvm/hai-qhsvm-pi5-tee-hai-tables-arXiv_2409.04935.md`, `docs/experimental-data/{accuracy,algorithms,device}.csv`.